Physics 4220H: Assignment II

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Problem 1. The intensity of sunlight hitting the earth is about $1300\,\mathrm{W\,m^{-2}}$. If sunlight strikes a perfect absorber, what pressure does it exert? What about a perfect reflector? Comment on the fraction of atmospheric pressure that the radiation pressure of sunlight amounts to?

Solution 1. Coming straight from the formula in the text,

$$P_{\text{radiation}} = \frac{1}{2}\epsilon_0 E_0^2 = \frac{I}{c}$$

which for us is around $4.34 \times 10^{-6}\,\mathrm{Pa}$ for a perfect absorber, and since prefect reflectors get both the momentum from the initial impact and the turn-around momentum from reflection a perfect reflector would see a pressure of around $8.67 \times 10^{-6}\,\mathrm{Pa}$. Atmospheric pressure is roughly $10 \times 10^5\,\mathrm{Pa}$, 1.18×10^{10}(!) times greater than the radiation pressure due to sunlight on a perfect reflector.

Problem 2. You might be confused by the fact that the intensity of light in a linear medium, $I = \epsilon_0 cnE_0^2$, scales with the index of refraction and thus it is greater in the material with the higher index. Consider Brewster's condition for TM light and compare the intensities of the incident, reflected, and transmitted (refracted) waves. Comment on the result.

Solution 2. At Brewster's angle all light is transmitted and since $T + R = 1$, $T = 1$ and so $I_I = I_T$, which is equivalent to saying that, for incident angle θ_B and transmitted angle θ_T

$$\frac{1}{2}\frac{n_1}{c\mu_1}E_{0I}^2\cos\theta_B = \frac{1}{2}\frac{n_2}{c\mu_2}E_{0T}^2\cos\theta_T.$$

Cancelling common factors and taking the approximation that $\mu_1 \approx \mu_2 \approx \mu_0$,

$$E_{0T}^2 = \frac{n_1}{n_2}E_{0I}^2\frac{\cos\theta_B}{\cos\theta_T}.$$

Since we are considering the case where $n_2 > n_1$, the ratio $n_1/n_2 < 1$ and under the same condition Snell's law tells us that $\theta_T < \theta_B$ which equates to, for cosine between 0 and $\pi/2$, $\cos\theta_T > \cos\theta_B$ and hence the ratio $\cos\theta_B/\cos\theta_T < 1$. These inequalities are preserved under a square root and so we can directly compare field amplitudes with

$$E_{0T} = abE_{0I}$$

where a and b are standins for the square roots of the ratios. My point is that both a and b are less than 1, and hence $E_{0T} < E_{0I}$, which is why the apparent issue with intensity is actually fine. The field scales down when entering an optically "denser"(?) medium (one of the higher IOR) proportional to the angle at which it enters and the relative IORs, preserving overall intensity.

Problem 3. Starting from the Fresnel equations for E_{0R}/E_{0I} and E_{0T}/E_{0I}, show that energy is conserved across the boundary, such that $T + R = 1$. Do this both for TM and TE incidence.

Solution 3. (a) **TM**: The Fresnel equations in this case tell us that

$$E_{0R} = \frac{\alpha - \beta}{\alpha + \beta} E_{0I} \quad \text{and} \quad E_{0T} = \frac{2}{\alpha + \beta} E_{0I}$$

for $\alpha = \cos\theta_T / \cos\theta_I$ and $\beta \approx n_2/n_1$. Since intensity is, in general, given by

$$I = \frac{1}{2}\epsilon v E_0^2$$

the Fresnel equations give intensities

$$I_I = \frac{1}{2}\epsilon_1 v_1 E_{0I}^2 \cos\theta_I, \quad I_R = \frac{1}{2}\epsilon_1 v_1 \left(\frac{\alpha - \beta}{\alpha + \beta}\right)^2 E_{0I}^2 \cos\theta_I, \quad I_T = \frac{1}{2}\epsilon_2 v_2 \left(\frac{2}{\alpha + \beta}\right)^2 E_{0I}^2 \cos\theta_T$$

where the cosines arise, as discussed in class, because we consider the amount of energy (or rather power) per unit area over the interface, not some plane at angle θ_I or θ_T to the interface-normal. Also note I've applied the law of reflected angles already. Now we consider (implicitly using the nonmagnetic approximation where appropriate)

$$\begin{aligned}
I_R + I_T &= \frac{E_{0I}^2}{2}\left[\epsilon_1 v_1 \left(\frac{\alpha - \beta}{\alpha + \beta}\right)^2 \cos\theta_I + \epsilon_2 v_2 \left(\frac{2}{\alpha + \beta}\right)^2 \cos\theta_T\right] \\
&= \frac{E_{0I}^2}{2c\mu_0}\left[n_1 \left(\frac{\alpha - \beta}{\alpha + \beta}\right)^2 \cos\theta_I + n_2 \left(\frac{2}{\alpha + \beta}\right)^2 \cos\theta_T\right] \\
&= \frac{E_{0I}^2}{2c\mu_0}\left[n_1 \cos\theta_I \frac{\alpha^2 - 2\alpha\beta + \beta^2}{\alpha^2 + 2\alpha\beta + \beta^2} + \frac{4 n_2 \cos\theta_T}{\alpha^2 + 2\alpha\beta + \beta^2}\right] \\
&= \frac{E_{0I}^2 n_1 \cos\theta_I}{2c\mu_0}\left[\frac{(n_1 \cos\theta_I)\left(\alpha^2 - 2\alpha\beta + \beta^2\right) + 4 n_2 \cos\theta_T}{\alpha^2 + 2\alpha\beta + \beta^2}\right] \\
&= \frac{E_{0I}^2 n_1 \cos\theta_I}{2c\mu_0}\left[\frac{\alpha^2 - 2\alpha\beta + \beta^2 + 4\alpha\beta}{\alpha^2 + 2\alpha\beta + \beta^2}\right] \qquad \text{Factor out } n_1 \cos\theta_I \\
&= \frac{E_{0I}^2 n_1 \cos\theta_I}{2c\mu_0}\left[\frac{\alpha^2 + 2\alpha\beta + \beta^2}{\alpha^2 + 2\alpha\beta + \beta^2}\right] 1 \\
&= \frac{1}{2}\frac{n_1}{2c\mu_0} E_{0I}^2 \cos\theta_I
\end{aligned}$$

and using $v = 1/\sqrt{\epsilon\mu_0}$ and $v = c/n$,

$$I_R + I_T = \frac{1}{2}\epsilon_1 v_1 E_{0I}^2 \cos\theta_I = I_I.$$

(b) **TE**: The Fresnel equations in this case tell us that

$$E_{0R} = \frac{1 - \alpha\beta}{1 + \alpha\beta} E_{0I} \quad \text{and} \quad E_{0T} = \frac{2}{1 + \alpha\beta} E_{0I}$$

for $\alpha = \cos\theta_T / \cos\theta_I$ and $\beta \approx n_2/n_1$. This gives us intensities

$$I_I = \frac{1}{2}\epsilon_1 v_1 E_{0I}^2 \cos\theta_I, \quad I_R = \frac{1}{2}\epsilon_1 v_1 \left(\frac{1 - \alpha\beta}{1 + \alpha\beta}\right)^2 E_{0I}^2 \cos\theta_I, \quad I_T = \frac{1}{2}\epsilon_2 v_2 \left(\frac{2}{1 + \alpha\beta}\right)^2 E_{0I}^2 \cos\theta_T.$$

Again we consider the sum of the reflected and transmitted intensities, using the nonmagnetic approximation

where appropriate

$$\begin{aligned}
I_R + I_T &= \frac{E_{0I}^2}{2c\mu_0}\left[n_1\left(\frac{1-\alpha\beta}{1+\alpha\beta}\right)^2\cos\theta_I + n_2\left(\frac{2}{1+\alpha\beta}\right)^2\cos\theta_T\right] \\
&= \frac{E_{0I}^2}{2c\mu_0}\left[n_1\cos\theta_I\frac{1-2\alpha\beta+(\alpha\beta)^2}{1+2\alpha\beta+(\alpha\beta)^2} + n_2\cos\theta_T\frac{4}{1+2\alpha\beta+(\alpha\beta)^2}\right] \\
&= \frac{E_{0I}^2}{2c\mu_0}\left[\frac{(n_1\cos\theta_I)\left(1-2\alpha\beta+(\alpha\beta)^2\right)+4n_2\cos\theta_T}{1+2\alpha\beta+(\alpha\beta)^2}\right] \\
&= \frac{E_{0I}^2 n_1\cos\theta_I}{2c\mu_0}\left[\frac{1-2\alpha\beta+(\alpha\beta)^2+4\alpha\beta}{1+2\alpha\beta+(\alpha\beta)^2}\right] \\
&= \frac{E_{0I}^2 n_1\cos\theta_I}{2c\mu_0}\left[\frac{1+2\alpha\beta+(\alpha\beta)^2}{1+2\alpha\beta+(\alpha\beta)^2}\right]1
\end{aligned}$$

which, reworking as before, yields

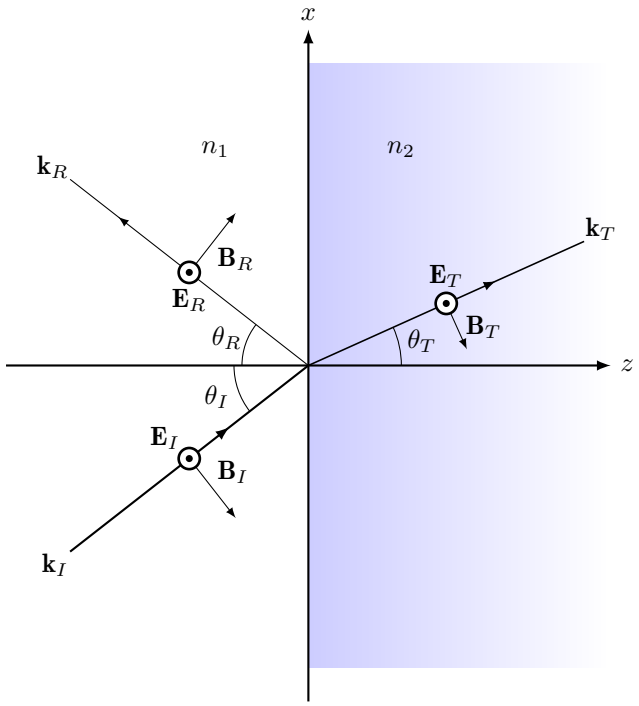
$$I_R + I_T = \frac{1}{2}\epsilon_1 v_1 E_{0I}^2\cos\theta_I = I_I.$$

Problem 4. Boundary conditions yielded the following "four" equations for the case of TM incidence:

$$\epsilon_1(-E_{0I}+E_{0R})\sin\theta_I = -\epsilon_2 E_{0T}\sin\theta_T \tag{1}$$
$$0 = 0\,(\text{No } B_z) \tag{2}$$
$$(E_{0I}+E_{0R})\cos\theta_I = E_{0T}\cos\theta_T \tag{3}$$
$$\frac{1}{\mu_1 v_1}(E_{0I}-E_{0R}) = \frac{1}{\mu_2 v_2}E_{0T}. \tag{4}$$

Draw a detailed diagram and derive the 4 equivalent equations for **TE** incidence.

Solution 4.



I am extremely proud of that diagram, now to actually solve the problem. I'll skip the first couple steps to obtain

the laws of reflection and refraction and instead focus on applying boundary conditions,

$$\epsilon_1(\mathbf{E}_{0I} + \mathbf{E}_{0R})_z = \epsilon_2(\mathbf{E}_{0T})_z \tag{5}$$

$$(\mathbf{B}_{0I} + \mathbf{B}_{0R})_z = (\mathbf{B})_z \tag{6}$$

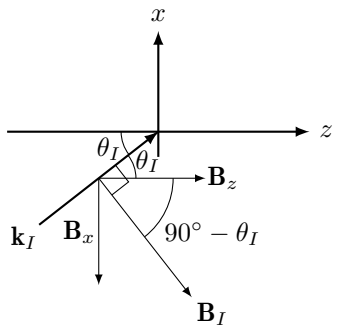
$$(\mathbf{E}_{0I} + \mathbf{E}_{0R})_{x,y} = (\mathbf{E}_{0T})_{x,y} \tag{7}$$

$$\frac{1}{\mu_1}(\mathbf{B}_{0I} + \mathbf{B}_{0R})_{x,y} = \frac{1}{\mu_2}(\mathbf{B}_{0T})_{x,y}. \tag{8}$$

Since we are in the transverse-electric case, there is no component of the electric field lying in $\hat{\mathbf{z}}$ and hence eq. (5) tells us nothing useful. Equation (6) tells us that

$$B_{0I}\sin(\theta_I) + B_{0R}\sin(\theta_R) = B_{0T}\sin\theta_T$$

where the angles come from



which is a bit confusing, but the process is to use the Z-pattern to identify the second θ_I, then note that $\mathbf{k}_I$ forms a right angle with $\mathbf{B}_I$ by definition and hence, because the angle between $\mathbf{B}_{Iz}$ and $\mathbf{k}_I$ is θ_I, the angle between $\mathbf{B}_{Iz}$ and $\mathbf{B}_I$ is $90° - \theta_I$ and so the projection onto z of $\mathbf{B}_I$ is $B_I\sin\theta_I$, with positive sign from the diagram. The process is similar for the reflected and transmitted components. Then using the law of refraction to say that $\theta_I = \theta_R$ and the relationship for field magnitudes to say that $E = Bv$, eq. (6) really tells us that

$$v_1\left(E_{0I} + E_{0R}\right)\sin(\theta_I) = v_2 E_{0T}\sin(\theta_T).$$

Equation (7) says that, since $\mathbf{E}$ is all in $\hat{\mathbf{y}}$ in all cases, that

$$E_{0I} + E_{0R} = E_{0T}.$$

Finally, eq. (8) says that, because $\mathbf{B}$ is only in $\hat{\mathbf{x}}$ and $\hat{\mathbf{z}}$, and using similar reasoning as to our work for eq. (6) to determine projections this time onto the $\hat{\mathbf{x}}$-axis we obtain

$$\frac{1}{\mu_1}(-B_{0I} + B_{0R})\cos\theta_I = -\frac{1}{\mu_2}B_{0T}\cos\theta_T \tag{9}$$

where I've implicitly applied $\theta_R = \theta_I$ and the signs on the I and T terms are a result of the diagram I drew. Transforming eq. (9) using $E = Bv$ and writing the four equations together we obtain

$$\begin{aligned}
&0 = 0\,(\text{No } E_z)\\
&\frac{1}{v_1}\left(E_{0I} + E_{0R}\right)\sin\theta_I = \frac{1}{v_2}E_{0T}\sin\theta_T\\
&E_{0I} + E_{0R} = E_{0T}\\
&\frac{1}{v_1\mu_1}(-E_{0I} + E_{0R})\cos\theta_I = -\frac{1}{v_2\mu_2}E_{0T}\cos\theta_T.
\end{aligned}$$

Problem 5. In lecture, we analyzed the expressions we obtained for Fresnel coefficients t, r, T, and R to draw detailed plots of these parameters in the case of external reflection (going from optically rare to optically dense media; $n_1 < n_2$). Do the same thing for the case of internal reflection (specifically for the case $n_1 = 1.5$, $n_2 = 1.3$; incidence from glass into water), by attending to the following questions:

(a) Is there a critical angle for TIR (If so, what is it)?

(b) Is there a Brewster's angle for which there is no reflection (If so, what is it)?

(c) Is it possible for either/both t and r to exceed 1?

(d) At what value are T and R equal?

(e) Put all these ideas together and plot/sketch two detailed graphs (one for TE and one for TM) for t, r, T, and R for internal reflection from glass into water.

(f) [Bonus] What can you say about the phase of the fields upon reflection in this case? Are there any π flips? If so, at what angle(s)?

Solution 5. I'll do this for TM incidence where

$$r = \frac{\alpha - \beta}{\alpha + \beta} \quad \text{and} \quad t = \frac{2}{\alpha + \beta}$$

and so $R = r^2$ and $T = \alpha\beta t^2$.

(a) Certainly, since $\beta = 1.3/1.5 = 13/15 < 1$ and hence $\arcsin(\beta)$ is defined and is equal to around $60°$. Any angle greater than this will have zero transmission.

(b) Since $\theta_B = \arctan(\beta)$, $\theta_B \approx 41°$.

(c) Yes for t, no for r. t actually starts higher than 1, at $0°$ $t = 2/\beta = 30/13$ and only continues to increase up to (nearly) θ_C, the critical angle we found in (a), where it takes the value

$$\frac{2n_1 \cos(\arcsin(13/15))}{n_1 \cos\left(\arcsin\left(\frac{13n_1}{15n_2}\right)\right) + n_2 \cos(\arcsin(13/15))} \approx 2.3.$$

For r, no since $r > 1$ would mean $R > 1$ since $R = r^2$ which would mean more energy is reflected than is incident. For transmission this is not a problem as $T = \alpha\beta t^2$ and so t can take a maximum of $1/\sqrt{\alpha\beta}$ before conservation of energy becomes a problem.

(d) Since $T + R = 1$, when $T = R = 1/2$.

(e) (Plots on the next page)

(f) I believe the only π phase flip occurs at θ_B for TM incidence, as that is the only time the reflected field's sign changes.

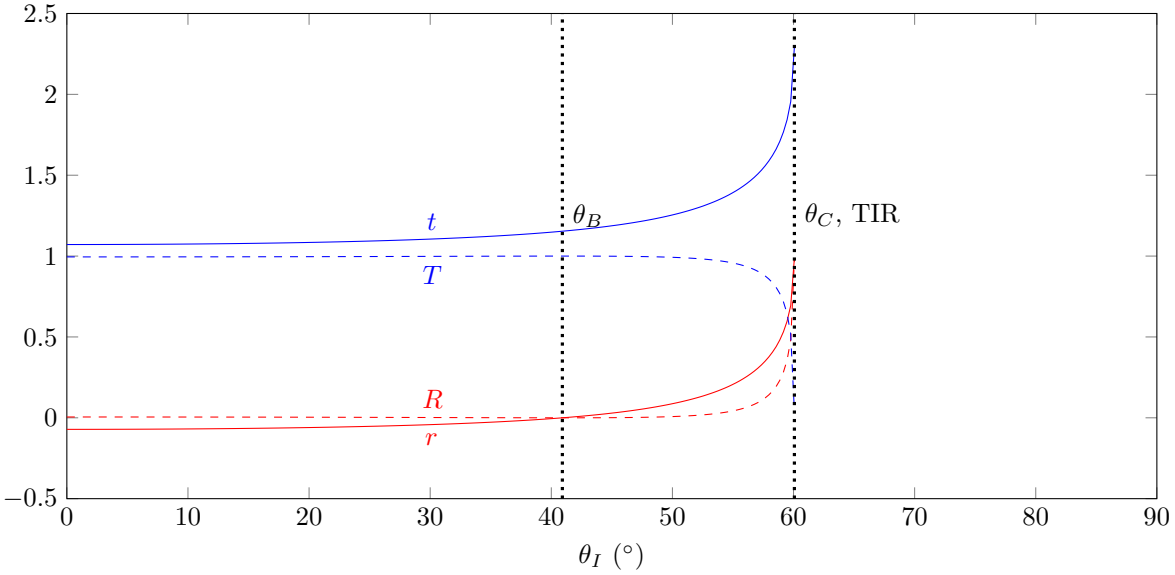


Figure 1: TM coefficients. Solid lines are r and t, dashed are R and T.

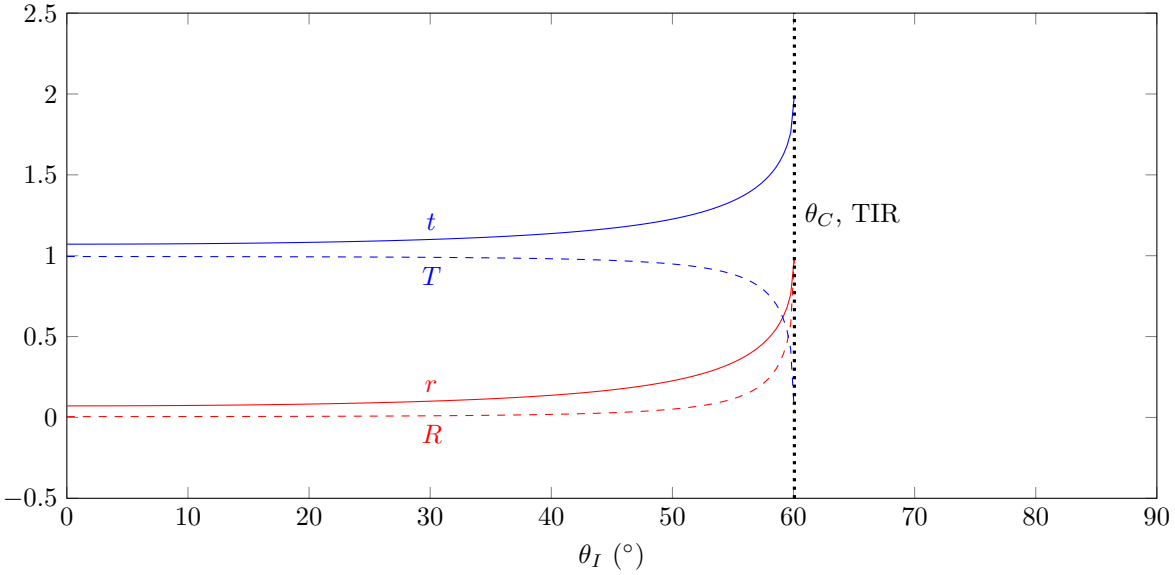


Figure 2: TE coefficients. Solid lines are r and t, dashed are R and T.